# Harnessing Keyness:
# Corpus-based Approach to ESP Material Development

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Concordancers often provide an option to generate lists of keywords. Keywords are the words that occur disproportionately more frequently in a particular text type (e.g. business English) compared to another text type (e.g. general English). This is one way of distinguishing technical or domain-specific words from general words. Novice users of concordancers tend to expect that the keyword lists produced are identical, yet there are significant differences in the lists generated. This paper shows how keyword lists are affected by the choices of concordancer, reference corpus and statistical test. ESP materials developers can use this knowledge to make a more informed choice of the variables so that the most appropriate keyword list for the target audience can be created.

## Introduction

The identification of words that deserve inclusion in teaching materials is a difficulty that many materials developers face. There are many factors to consider in the selection of vocabulary, such as frequency, appropriacy, expediency, need and level. The most frequent words in a text are relatively easy to identify, but are not necessarily the most useful words to highlight in ESP materials. This is because grammatical words and high frequency general words are likely to occupy the top positions. Words that are key, however, are likely to merit inclusion. Concordancers can be harnessed to identify the frequency and keyness of vocabulary.

Simply put, keyness is a measure of the frequency with which a word occurs disproportionately in a particular text type. Keyness is assessed by comparing the relative frequency of a word in a focus corpus to a reference corpus using a statistical test. Words that are key are called keywords (Scott, 1997).

Novice users may expect all concordancers to produce the same keyword list for a text. However, this is not the case. Different concordancers, reference corpora and statistical tests result in radically different keyword lists.

Concordancers can be classified into four generations (McEnery and Hardie, 2012) although the first two generations are now obsolete. Fourth generation concordancers can deal with large corpora and are far more powerful than third generation concordancers (Table 1). Some concordancers provide options to upload a reference corpus to which the focus corpus can be compared while others provide a range of corpora from which the user can select. Concordancers may have a default statistical test (e.g. chi-squared in AntConc) or provide alternatives for the user to select from. Keyword list generation is underpinned by comparing the ratios of words occurring in the focus and reference corpora using statistical tests. Kilgarriff (2012, p.5) highlights two statistical problems. First, the resolution of dividing by zero when there are no occurrences of a word in the reference corpus. Second, overcoming the domination of words which occur rarely in the reference corpus. Different tests use different methods to address these issues.

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This paper explores how the choice of concordancer, reference corpus and statistical test generates different lists of keywords. Materials developers can use this knowledge to make more informed choices of which vocabulary to focus on in their tailor-made materials.

## Method

A corpus of texts comprising all the research articles published in the International Business Review (IBR) from February 2010 to October 2013 was manually collected and concatenated into a single text file. Table 2 shows the composition of this focus corpus.

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Insert Table 2 about here
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The three variables (concordancers, reference corpora and statistical tests) were each tested in turn. A popular third generation concordancer, AntConc 3.2.4w (Anthony, 2012) and a popular fourth generation concordancer, Sketch Engine (Kilgarriff et al., 2014), were selected for comparison. The raw frequency word count for each concordancer was first calculated. Keyword lists were generated using the British Academic Written English (BAWE) corpus and the Brown corpus in Sketch Engine (Table 3). A keyword list was then generated using the Brown corpus in AntConc. This was undertaken using three different statistical tests in Sketch Engine and two different tests in AntConc. The keyword lists were then evaluated from the perspective of an ESP materials developer.

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## Findings

Findings regarding each of the three variables are described, interpreted and evaluated in the following sections.

<u>Concordancers</u>

The raw count of frequency of words in both AntConc and Sketch Engine results in the same order for the top ten words, yet only the word count for *that* is identical (Table 4). This raw word count difference can be accounted for by differences in the operational definition of a word and the process of tokenization. Anthony (2013) notes that Wordsmith Tools and AntConc count contractions differently, e.g. *we'll* is counted as one word in Wordsmith, but two words in AntConc. Word count is just one variable in the calculation of keyness. Since results differ at the level of raw word count, this difference may be exacerbated when other variables are added.

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Each concordancer offers different functionality with regard to calculating keyness. For example, AntConc allows users to upload their own reference corpus and provides the

choice of either chi-squared or log-likelihood for the statistical test while Sketch Engine subscription incorporates access to numerous reference corpora and 4 different statistical tests. For most material developers, the functionality of the concordancer is most likely of more importance than a thorough understanding of the definition of words and tokenization process used.

Reference corpora

Table 5 shows the keyword lists created in Sketch Engine using the same statistical test (Midway) but with difference reference corpora. Keyword lists created when using the BAWE corpus and Brown corpus shared five of the top ten results. The remaining five words in BAWE appeared more specialized than the Brown corpus. The BAWE keyword list, therefore, appears more appropriate for learners with a stronger vocabulary base.

Scott (2009) claims that there is no bad reference corpus. However, different reference corpora yield radically different keyword lists. The genre and diachrony of a corpus are found to significantly affect keyness (Goh, 2010). Given that different reference corpora impact the generated keyword lists, materials developers would be well advised to compare the results using different reference corpora.

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Statistical tests

As shown in Table 6, selecting the log-likelihood and chi-squared tests in AntConc using the Brown corpus resulted in identical lists for the first eight keywords. Simple ratios, such as log-likelihood and chi-squared, (Kilgarriff, 2012, p.5) produce keyword lists "dominated by rare words". Gabrielatos and Marchi (2012) oppose the use of log-likelihood and chi-squared to calculate keyness due to frequency bias and assumptions on the random nature of language.

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Insert Table 6 about here
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Table 7 shows the keyword lists generated in Sketch Engine using the BAWE Corpus, but selecting different statistical tests. The simple maths version (Kilgarriff, 2009) in Sketch Engine names the tests clearly (e.g. Common, Rare) and is not based on the assumption that language is random (Kilgarriff, 2005). Rare resulted in higher occurrence of rare words while Common resulted in a skew to more common words. When selecting vocabulary for less proficient students, it may be prudent to use a keyword list generated using Common.

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Insert Table 7 about here
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**Conclusion**

The three variables of concordancer, reference corpora and statistical tests greatly affect the keyword lists generated. Although AntConc has many advantages particularly in classroom-based data-driven learning, fourth generation concordancers that can deal with larger corpora and provide reference corpora could save materials developers a great deal of time. Sketch Engine provides an easy, quick and affordable way to calculate a variety of

keyword lists. The availability of 20 reference corpora and 4 appropriately-named statistical tests make it easy to tailor keyword lists to the intended learners. Selecting a general English reference corpus and the Common statistical test in Sketch Engine is likely to generate keyword lists that are more suitable for lower level students.

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**Biodata**

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**Table 1**
***Current Generations of Concordancers***

| | 3rd generation | 4th generation |
|---|---|---|
| **Location** | personal computers | web servers |
| **Size of corpora** | Small corpora - low millions | Large corpora – 100 million+ |
| **Examples** | AntConc (Anthony, 2012) | CQPweb (Hardie, 2013) |
| | UAM Corpus Tool (O`Donnell, 2013) | Sketch Engine (Kilgariff et al., 2014) |
| | Wordsmith Tools (Scott, 2012) | W-matrix (Rayson, 2008) |

**Table 2**
***IBR Focus Corpus***

| | Count (made in AntConc 3.2.4w) |
|---|---|
| Tokens | 2,516,051 |
| Words | 1,966,650 |
| Sentences | 77,547 |

**Table 3**
***Outline of Reference Corpora Used***

| | BAWE corpus | Brown corpus |
|---|---|---|
| Date created | 2000s | 1960s |
| Type of corpus | Academic | General |
| Type of English | British | American |
| Words | 6,506,995 | 1,000,000 |

**Table 4**
***Raw Frequency Results***

| | | Sketch Engine | AntConc |
|---|---|---|---|
| **1** | the | 106,022 | 106,064 |
| **2** | and | 77,508 | 77,542 |
| **3** | of | 72,733 | 72,990 |
| **4** | to | 47,454 | 47,834 |
| **5** | in | 41,791 | 42,056 |
| **6** | a | 32,007 | 32,336 |
| **7** | that | 23,092 | 23,092 |
| **8** | is | 21,249 | 21,245 |
| **9** | for | 17,293 | 17,303 |
| **10** | as | 14,309 | 14,329 |

**Table 5**
***Keyword Lists using BAWE and Brown in Sketch Engine with Midway Test***

| | BAWE | Brown |
|---|---|---|
| **1** | firms | firms |
| **2** | firm | firm |
| **3** | export | export |
| **4** | foreign | Table |
| **5** | subsidiary | variables |
| **6** | internationalization | international |
| **7** | FDI | markets |
| **8** | subsidiaries | knowledge |
| **9** | markets | foreign |
| **10** | MNEs | market |

**Table 6**
***Keyword Lists using Log-likelihood and Chi-squared Tests in AntConc with Brown Corpus***

| | Log-likelihood | Chi-squared |
|---|---|---|
| **1** | the | the |
| **2** | firms | firms |
| **3** | firm | firm |
| **4** | al | et |
| **5** | et | al |
| **6** | in | In |
| **7** | knowledge | knowledge |
| **8** | market | market |
| **9** | this | international |
| **10** | table | foreign |

**Table 7**
***Keyword Lists using Three Statistical Tests in Sketch Engine with BAWE Corpus***

| | Rare | Midway | Common |
|---|---|---|---|
| **1** | OFDI | firms | and |
| **2** | offshoring | firm | firms |
| **3** | Vahlne | export | firm |
| **4** | multinationality | foreign | foreign |
| **5** | Full-size | subsidiary | knowledge |
| **6** | MathML | internationalization | international |
| **7** | Kogut | FDI | market |
| **8** | BOP | subsidiaries | country |
| **9** | MathJax | markets | Table |
| **10** | Ghoshal | MNEs | performance |